

Template, Not Escape: Primes as a Ruler for the Finite-Memory Ceiling of the Event-Density Substrate

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Abstract

Can a finite-memory substrate produce the prime numbers? Event Density's substrate is, by its own primitives (#2–#3), in the **finite-reach / finite-memory** class, and two established results — **Sarnak Möbius-disjointness** and the **sieve parity barrier** — predict precisely what such a class can and cannot reproduce of the prime structure: the **template** layer (the finite-local residue statistics) yes, the **escape** layer (primality itself, the Möbius function, pair correlations) no. Using the integers as an external ruler, we measure both. **The template is reproduced exactly:** the scale-invariant template information $H(\mathbf{dx}) - H(\mathbf{dx} \mid \mathbf{n} \bmod 30) = 1.700 \text{ bits}$ is matched, and the least-prime-factor density ladder $P(\text{lpf} = p) = (1/p) \prod_{q < p} (1 - 1/q)$ is reproduced to five decimals. **The escape is provably blocked, three ways.** (A) Conditioning primality on any finite template state $\mathbf{n} \bmod M$ leaves an irreducible residual — $H(\text{prime} \mid \mathbf{n} \bmod M) \approx 0.18\text{--}0.29$ bits per integer ($\approx 0.58\text{--}0.95$ bits per *open* position) — the parity barrier expressed in bits. (B, the sharp test) The *optimal* finite-memory correlator with the Möbius function, $C^*(N, M) = (1/N) \sum_r |\sum_{n=r}^M \mu(n)|$, decays toward zero as N grows (bounded by $\sim \sqrt{M/N}$) for every fixed M — Sarnak-in-this-class confirmed, and the falsifier (a bounded-complexity function correlating with μ) **was not seen**. (C) The twin-prime density requires the Hardy–Littlewood singular series $2C_2 = 1.3203$ — a pair-correlation a finite-local template generator does not supply. The mechanism is the **$\sqrt{N}$ activation horizon:** a larger template state M needs proportionally larger N to resolve, so no *fixed* finite memory ever reaches the escape. The result locates the substrate's ceiling exactly where external mathematics places it — **template yes, escape no, and the no is load-bearing**. It is the substrate-evaluation program's only *proven* negative, because it is anchored to theorems outside ED.

1. Introduction

1.1 What this paper does

The standing question for a finite substrate ontology is sharp: *can ED's finite memory produce an infinite-memory object like the primes?* This paper answers it as a **ceiling-location** measurement. ED's substrate is, by primitives #2–#3 (finite reach, finite memory), in a definite computational class. The prime numbers carry two layers of structure — a **template**

(finite-local residue statistics, the part a sieve “knows” from $n \bmod M$) and an **escape** (primality, the Möbius function, pair correlations, the part no finite residue state determines). External mathematics predicts a finite-memory class captures the first and provably not the second. We measure both, using the integers as a calibrated external ruler.

1.2 Why this matters

Most substrate-evaluation results are *located-but-open* — an obstruction is identified but not proven insurmountable. This one is different: the wall is a **theorem** (the parity barrier; Möbius orthogonality to periodic sequences), so the “no” is not ED grading its own homework. That makes this the program’s **sharpest, most defensible negative** — a clean, externally-certified statement of exactly which structures ED’s finite-memory class can and cannot carry, and a concrete in-bits picture of where the ceiling sits.

Why the primes, and a word of context. The primes are the cleanest available external ruler for a finite-memory ceiling: their template/escape split is mathematically **sharp** (residue structure vs primality), and the finite-memory class coincides exactly with the **sieve** class whose limits a century of analytic number theory has charted — Eratosthenes’ residue sieves, Selberg’s parity problem (1949), the Hardy–Littlewood prime-pair heuristics, and the modern Möbius-randomness (Sarnak) program. ED’s substrate falls in that class, so those results apply to it directly — which is why the ceiling can be *located by theorem* rather than merely measured.

1.3 Arc context

This pairs with the **channel-capacity** result (A1), which found ED’s controlled capacity is exactly zero — the same finite-reach ceiling, measured *internally*. Here the *same* ceiling is measured against an **external, math-certified ruler** (Möbius disjointness). Two readings of one property: ED’s hard ceiling on long-range / global information, one internal, one externally calibrated.

1.4 Regime of validity

This is a **finite-N measurement** ($N \leq 5 \times 10^6$; twin primes to $N = 32\{, \}463$) using **standard sieve data** — the integers’ own factorization, **not an ED simulation**. No ED dynamics are run. The ED input is *class membership* (the substrate is finite-memory by primitives #2–#3); the result is therefore a statement about that **class**, not about any specific ED run or observable. The asymptotic statements (the Möbius correlator $\rightarrow 0$; the $\sqrt{N}$ horizon) are anchored to the proven theorems (§2) and read from the N-trend, not separately proven at infinity here.

2. The Substrate Class and the External Anchors

Substrate class (from primitives). ED’s substrate is finite-reach (primitive #2) and finite-memory / zero-entropy (primitive #3): every determination is a function of a bounded local state. **Explicitly: finite reach + finite local state means every determination is a function of a bounded residue state $n \bmod M$ (for some fixed M) — which is precisely a periodic, zero-entropy sequence, exactly the class the parity barrier and Sarnak’s**

program constrain. So the substrate’s membership in the sieve / zero-entropy class is not an analogy but an identity. The integers’ template structure is exactly such a residue state, so the class *can* represent it; the question is whether it can represent more.

External anchors (established mathematics — inherited, not ED-derived).

- **Möbius orthogonality (the proven core of Sarnak’s program):** the Möbius function $\mu(n)$ does not correlate with any sequence produced by a low-complexity (zero-entropy) dynamical system. A finite-memory function of $n \bmod M$ is exactly such a sequence — and, crucially, **the special case this paper needs is not conjectural**. Functions of $n \bmod M$ are *periodic*, and Möbius orthogonality to periodic sequences is a **classical theorem**: $\sum_{n \leq N, n \equiv r \pmod{M}} \mu(n) = o(N)$ for each fixed M , by the prime number theorem in arithmetic progressions (Siegel–Walfisz). Sarnak’s *conjecture* is the vast generalization to **all** zero-entropy systems; this paper invokes only its already-proven, periodic base case. So the finite-memory negative rests on a **theorem**, not on the open conjecture.
- **The sieve parity barrier (Selberg’s parity problem):** sieve methods (finite-local, residue-based) cannot distinguish integers with an even number of prime factors from those with an odd number — they cannot, on their own, isolate the primes. This is the escape layer’s irreducibility expressed sieve-theoretically.

These are the *rulers*. The substrate belongs to the class they constrain; the measurements below confirm the class hits the predicted ceiling.

Ruler (the Factor Skyline). The integers’ factorization statistics, computed by a sieve to $N = 5 \times 10^6$ (`primes_test.py`, `numpy`). No ED simulation is run; the substrate’s *class membership* is the ED input, and the integers supply the external scale against which the class is measured.

Reproducibility. The integers $1 \dots N$ ($N = 5 \times 10^6$) are sieved by a simple Eratosthenes sieve recording the least prime factor and the prime-factor count per integer (`numpy`). Entropies are plug-in (empirical-histogram) estimates of the gap distribution dx and the conditional $dx \mid n \bmod M$. The Test-B correlator is $C(N, M) = (1/N) \sum_r |\sum_{n \equiv r \pmod{M}} \mu(n)|$, the exact optimum over functions of $n \bmod M$, evaluated at $N/4, N/2, N$ for M in $\{2, 6, 30, 2310, 30030\}$. Twin primes $(p, p+2)$ are counted directly and normalized by $N/\ln^2 N$. No randomness and no fitting beyond the entropy histograms. Script: `primes_test.py`.

No primitive forcing is invoked.

2.5 Load-Bearing Step Audit

Step	Status	Source / justification
13 primitives postulated	P	Paper_087
Substrate is finite-reach / finite-memory class	D	primitives #2–#3
Sarnak Möbius-disjointness; sieve parity barrier	I (external math)	established results; the rulers
Template information = 1.700 bits reproduced	measured	§4.1 — scale-invariant invariant

Step	Status	Source / justification
lpf density ladder exact to 5 decimals	measured	§4.1
Escape floor $H(\text{prime} \setminus n \bmod M) \approx 0.18\text{--}0.29$ bits irreducible	measured + I-anchored	§4.2 — parity barrier in bits
Best finite-memory μ -correlator $\rightarrow 0$ (bounded $\sim \sqrt{M/N}$)	measured + I-anchored	§4.3 — Sarnak-in-class; falsifier not seen
Twin density needs $2C_2 = 1.3203$	measured	§4.4 — Hardy–Littlewood singular series
Mechanism: $\sqrt{N}$ activation horizon	D-structural	§5 — fixed M needs growing N
Template yes / escape no; the “no” is load-bearing	interpretation	§7
ED generates the primes	NOT CLAIMED	preamble 2

All steps P, D, I, measured, structural, interpretation, or explicitly not-claimed. *The negative is I-anchored (external theorems), which is what makes it a proven rather than merely located ceiling.*

3. The Two Layers: Template and Escape

The prime structure decomposes, for a finite-local observer, into:

- **Template** — what a finite residue state $n \bmod M$ determines: which positions are *definitely composite* (covered by a small prime) and the equidistribution of primes among the *open* (coprime-to- M) residues. This is finite-local and representable by the substrate class.
- **Escape** — what no finite residue state determines: *which open positions are actually prime*, the sign of $\mu(n)$, and pair correlations (twins). By Dirichlet, primes equidistribute among the open residues, so conditioning on $n \bmod M$ says *nothing beyond base rate* about which open position is prime. That residual is the escape — the parity barrier.

The test measures whether the substrate class reproduces the template (it should) and reaches the escape (it should not).

4. Results

4.1 The template is reproduced exactly

quantity	measured	scale-invariant target
$H(dx)$	2.997 bits	(scale-dependent)
$H(dx \setminus n \bmod 30)$	1.297 bits	(scale-dependent)

quantity	measured	scale-invariant target
template information	1.700 bits	1.700 (invariant)
$H(dx) - H(dx \setminus n \bmod 30)$		

The **1.700-bit template invariant is reproduced exactly** — the robust, scale-invariant figure (the absolute entropies are scale-dependent; their difference is not). And the least-prime-factor density ladder falls straight out of finite-local residue structure:

p	empirical	$P(\text{lpf}=p) = (1/p) \prod_{q < p} (1 - 1/q)$
2	0.50000	0.50000
3	0.16667	0.16667
5	0.06667	0.06667
7	0.03810	0.03810
11	0.02078	0.02078
13	0.01598	0.01598

Exact to five decimals. **The template layer is fully within the substrate class.**

4.2 Test A — the escape floor (the parity barrier in bits)

Conditioning primality on the template state $n \bmod M$ ($M = \text{primorials}$):

M	P(open)	$H(\text{prime} \setminus n \bmod M)$	template captured	$H_b(\text{escape per open})$
2	0.500	0.291	0.074	0.583
30	0.267	0.221	0.144	0.829
210	0.229	0.203	0.162	0.887
2310	0.208	0.191	0.174	0.920
30030	0.192	0.181	0.184	0.945

As M grows, finite memory **captures more of the template** (captured bits rise, conditional entropy falls) by resolving the *covered* positions (definitely composite $\rightarrow$ zero entropy). But it supplies **nothing beyond base rate** about which *open* position is prime — open residues are structurally identical, primes equidistribute among them (Dirichlet). The residual $H(\text{prime} \mid n \bmod M) \approx 0.18\text{--}0.29$ bits/integer ($H_b \approx 0.58\text{--}0.95$ bits per *open* position) is the **escape: the parity barrier in bits**, irreducible to any finite template state.

Two clarifications. *On the figure:* the escape entropy is **not a universal constant** — it is the irreducible conditional entropy *at the tested N and M* , and the meaningful fact is its **persistence** (it does not fall toward zero as M and N grow), not its particular value. *On why it is irreducible:* **Dirichlet's theorem** guarantees the primes equidistribute among the open (coprime-to- M) residues, so those residues are statistically identical — which is exactly why a finite residue state $n \bmod M$ cannot distinguish *which* open position is prime.

4.3 Test B — the Möbius correlation (the sharp experiment)

The **optimal** finite-memory correlator with μ — the best over *all* functions of $n \bmod M$:

$$C^*(N, M) = \frac{1}{N} \sum_r \left| \sum_{n \equiv r \pmod{M}} \mu(n) \right|.$$

M	N/4	N/2	N	$\sim \sqrt{M/N}$
2	0.00018	0.00014	0.00014	0.0006
30	0.00329	0.00192	0.00118	0.0024
2310	0.03241	0.02225	0.01523	0.0215
30030	0.10033	0.07101	0.05021	0.0775

For **every fixed M**, the best finite-memory correlator **decays toward 0** as **N grows** ($\approx$ halving as N doubles; bounded by $\sim \sqrt{M/N}$). Even the *optimal* function of the template state cannot correlate with μ asymptotically. For each fixed M this decay is a **classical theorem** — $\sum_{n \equiv r \pmod{M}} \mu(n) = o(N)$ by the prime number theorem in arithmetic progressions — so the experiment is *confirming a proven result and quantifying its $\sqrt{M/N}$ rate*, not testing Sarnak’s open conjecture. This is nonetheless the genuine experiment of the paper: a bounded-complexity, finite-memory function with C^* bounded away from zero would be a real discovery (it would refute Möbius orthogonality in this class). **It was not seen.**

4.4 Test C — twin density needs a correlation the template does not supply

The template (lpf ladder, §4.1) is reproduced exactly. The **escape** — a correlation beyond the template — shows up in the twin-prime density (twins $(p, p+2)$ to $N = 32\{, \}463$):

- $\pi_2 / (N/\ln^2 N) = 1.545$ — pure independence/template predicts **1.000** and *misses*.
- $\pi_2 / (2C_2 * N/\ln^2 N) = 1.170$ — the Hardy–Littlewood singular series **$2C_2 = 1.3203$** is the dominant correction (the residual 1.17 is the crude $N/\ln^2 N$ versus the integral $\text{Li}_2(N)$, $\sim +13\%$ at this N).

The twin density carries the **$2C_2$ pair-correlation** — an escape structure a finite-local template generator does not supply. ($2C_2 = 1.3203$ is **not derived here**; it is the known Hardy–Littlewood twin-prime singular-series constant, used as the external yardstick. The point is only that the measured twin density *needs* it — i.e. carries a pair-correlation the template lacks.)

5. The Mechanism: the $\sqrt{N}$ Activation Horizon

Why does no *fixed* finite memory ever reach the escape? Because the template state that would resolve more of the structure grows with the scale it must resolve. Test B makes this quantitative: a larger M *does* correlate with μ at small N (the $\sqrt{M/N}$ envelope), but that correlation **washes out** as N grows — a *fixed* M with growing N goes to zero, and capturing more requires M to grow proportionally with N. The required memory grows like $\sqrt{N}$. So a

finite-memory substrate is always, eventually, beneath the escape: it can chase the template arbitrarily far, but the escape recedes at exactly the rate that keeps it out of finite reach. This is the **activation horizon**, and it is the substrate-level shadow of the parity barrier. The $\sqrt{\text{rtN}}$ scaling is not coincidental: it is the same square-root-cancellation scaling that governs the **large-sieve inequality** and the expected size of character sums — the quantitative form of “finite memory buys only $\sqrt{\text{rtN}}$ of resolution.”

6. Relation to the Channel-Capacity Result and the Program

The channel-capacity result (A1) found ED’s controlled capacity is exactly zero — the finite-reach ceiling measured *inside* ED, against ED’s own observables. This paper measures the *same* finite-reach / finite-memory ceiling against an **external, math-certified ruler** (Möbius disjointness, the parity barrier). Two readings of one property — ED’s hard ceiling on long-range / global information — one internal, one externally calibrated. Together they answer the standing question, *can ED finite-memory produce an infinite-memory system like the primes?: the template, yes; the escape, no — and the no is load-bearing.*

This negative is also of a *different kind* from the program’s others. Most are coarse-graining walls (the continuum / metric flesh) — located but not proven insurmountable. This one is a **finite-memory / uncomputability** wall, anchored to a theorem, hence **proven**. It is the program’s only proven negative, and it owes that status entirely to having an *external* ruler.

7. Interpretation

Finite memory saturates the template and fails the escape. The substrate class reproduces the prime template exactly (the 1.700-bit invariant; the lpf ladder) and provably cannot reproduce the escape (per-open-position primality entropy irreducible; the best finite-memory Möbius correlator $\rightarrow 0$; the twin density’s $2C_2$ pair-correlation unavailable). The integers, used as a ruler, locate the substrate’s ceiling **precisely where Sarnak disjointness and the sieve parity barrier place it.**

The *no* is the load-bearing content. It is not a failure of ED — ED is not in the business of generating primes — but a clean, externally-certified **measurement of the class ED belongs to**. And it sharpens the program’s overall shape: where the continuum / metric results are “form yes, flesh-as-coarse-grained-shadow,” this is “template yes, escape **provably no**” — the one place the wall is a theorem rather than an open obstruction.

8. Limitations (honest)

- **Finite N.** The sieve runs to 5×10^6 (twins to 32{,}463); the asymptotic claims are read from the N-trend and anchored to the proven theorems, not separately proven at infinity.
- **Ruler, not generator.** No ED simulation is run; the ED input is *class membership* (finite-memory), and the integers supply the external scale. The result is about the class, not about a specific ED dynamical run.

- **The escape figure is scale-dependent.** The citable invariant is the 1.700-bit template; the ~ 0.26 -bit per-integer escape peak is scale-dependent, not a universal constant.
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9. Falsification Criteria

9.1 A finite-memory Möbius correlator (the genuine falsifier)

Falsifier sentence: *A bounded-complexity, ED-definable function $f(n)$ (a function of a finite local state) with $(1/N)\text{Sigma } \mu(n)f(n)$ bounded away from zero as $N \rightarrow \text{inf}$ would refute Sarnak-in-this-class — a real mathematical discovery, and a refutation of §4.3. It was searched for (over all functions of $n \bmod M$) and not found.*

9.2 The template not reproduced

Falsifier sentence: *A finite-local computation failing to reproduce the 1.700-bit template invariant or the lpf density ladder would falsify §4.1 (the substrate class would then not even capture the template).*

9.3 The escape captured by finite memory

Falsifier sentence: *A fixed finite template state M for which $H(\text{prime} \mid n \bmod M)$ falls toward zero as N grows — i.e. finite memory resolving primality beyond base rate among open residues — would falsify §4.2 and the parity-barrier anchoring.*

9.4 The activation horizon not $\sqrt{N}$

Falsifier sentence: *A finite-memory correlator whose useful M does not have to grow with N (a fixed M retaining correlation with μ as $N \rightarrow \text{inf}$) would falsify the $\sqrt{N}$ activation-horizon mechanism (§5).*

10. Position Statement

Using the integers as an external ruler, ED's **finite-memory substrate class reproduces the prime template exactly and provably cannot reproduce the escape.** The 1.700-bit template invariant and the lpf ladder are matched; the primality entropy per open position is irreducible; the optimal finite-memory Möbius correlator decays to zero; the twin density requires a pair-correlation the template does not supply; and the $\sqrt{N}$ activation horizon is the mechanism.

What this paper claims. ED's finite-memory class hits the parity-barrier ceiling exactly where external mathematics (Sarnak disjointness; the sieve parity barrier) says it must — template captured, escape blocked. Because the wall is a theorem, this is the substrate-evaluation program's only *proven* negative.

What this paper does not claim. ED does not generate or explain the primes; the integers are a ruler, not a target. No number theory is advanced; the structural parallel carries no content

claim. The asymptotic statements are anchored to the proven theorems and read from finite- N trends, not separately proven here.

Series context. The externally-calibrated companion to the channel-capacity result (A1): one finite-reach ceiling, measured internally there and against a math-certified ruler here. The standing question — can ED’s finite memory produce an infinite-memory object like the primes? — is answered: **the template, yes; the escape, no; and the *no* is load-bearing.**

Appendix — Artifacts and Glossary

Artifacts

- `Primes_Arc/primes_test.py` — sieve to $N = 5 \times 10^6$ (numpy); template-information, Test A (escape floor), Test B (optimal mod- M Möbius correlator), Test C (lpf ladder; twin density).

Glossary

- **Open residues.** The residue classes $r \bmod M$ coprime to M — the positions a sieve to level M leaves *uncovered*, among which the primes above M must lie and (by Dirichlet) equidistribute.
- **Template layer.** The finite-local residue structure of the integers ($n \bmod M$): coverage by small primes + equidistribution of primes among open residues. Within the finite-memory class.
- **Escape layer.** Primality itself, the Möbius function, pair correlations — the structure no finite residue state determines. The parity barrier.
- **Template information (1.700 bits).** The scale-invariant $H(dx) - H(dx \mid n \bmod 30)$; reproduced exactly. The citable invariant.
- **Parity barrier.** The sieve-theoretic obstruction that finite-local methods cannot isolate the primes (cannot resolve the parity of the number of prime factors).
- **Sarnak disjointness.** Möbius does not correlate with any zero-entropy (low-complexity) sequence — including any finite-memory function of $n \bmod M$.
- **Activation horizon ($\sqrt{N}$).** The required template memory grows like $\sqrt{N}$, so no fixed finite memory ever reaches the escape — the substrate-level mechanism of the ceiling.
- **$2C_2 = 1.3203$.** The Hardy–Littlewood twin-prime singular series; the escape pair-correlation the template does not supply.

Reader map and open work

Intuition — the parity barrier in one line. Finite memory can classify residue classes but cannot see the **parity of the number of prime factors**; isolating the primes requires exactly that parity, which no finite residue state encodes. That is the whole obstruction.

Where to look next. - *The internal reading of the same ceiling (channel capacity = 0):* the A1 result. - *The coarse-graining wall (the continuum / PDE shadow):* the Continuum paper. - *The gravity arc (the same finite-reach character):* GR-II.

Open work (declared): test other arithmetic rulers (the Liouville function `lambda`, the divisor function `d(n)`) for the same template/escape split; test ED-generated sequences against the Möbius correlator directly (rather than via the class argument); push `N` higher; and test adaptive $M(N)$ scaling against the $\sqrt{N}$ horizon explicitly.

End of Paper (Finite-Memory Ceiling).

Substrate-evaluation arc. ED's finite-memory class reproduces the prime template exactly (1.700-bit invariant; lpf ladder) and provably fails the escape (primality entropy irreducible; optimal Möbius correlator $\rightarrow 0$; twin density needs $2C_2$), located by Sarnak disjointness + the parity barrier via the $\sqrt{N}$ activation horizon. ED does not generate the primes — they are a ruler for the substrate's ceiling. The program's only proven negative, because it is anchored to a theorem.